

# *Syntax -CKY, PCFGs*

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Week 5: Lecture 3

# CKY Algorithm

- Let  $n$  be the number of words in the input. Think about  $n + 1$  lines separating them, numbered 0 to  $n$ .
- $x_{ij}$  will denote the words between line  $i$  and  $j$
- We build a table so that  $x_{ij}$  contains all the possible non-terminal spanning for words between line  $i$  and  $j$ .
- We build the Table bottom-up.

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## Home Exercise

Use CKY algorithm to find the parse tree for “Book the flight through Houston” using the CNF form shown in the previous slide.

# CKY for CFG

| a<br>1 | pilot<br>2 | likes<br>3 | flying<br>4 | planes<br>5 |
|--------|------------|------------|-------------|-------------|
|        |            |            |             |             |
|        |            |            |             |             |
|        |            |            |             |             |
|        |            |            |             |             |
|        |            |            |             |             |

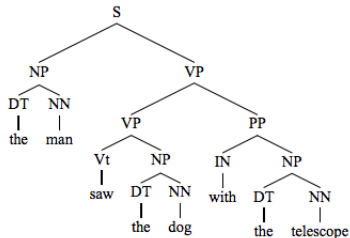
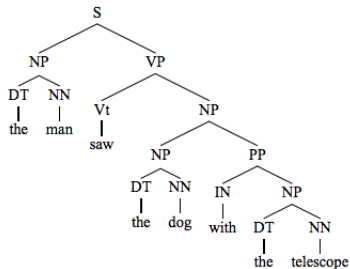
$S \rightarrow NP VP$   
 $VP \rightarrow VBG NNS$   
 $VP \rightarrow VBZ VP$   
 $VP \rightarrow VBZ NP$   
 $NP \rightarrow DT NN$   
 $NP \rightarrow JJ NNS$   
 $DT \rightarrow a$   
 $NN \rightarrow pilot$   
 $VBZ \rightarrow likes$   
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|--------|------------|------------|-------------|-------------|
| DT     | NP         | -          | -           | S S         |
|        | NN         | -          | -           | -           |
|        |            | VBZ        | -           | VP<br>VP    |
|        |            |            | JJ<br>VBG   | NP<br>VP    |
|        |            |            |             | NNS         |

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# What about Ambiguities?



# Probabilistic Context-free grammars (PCFGs)

PCFG:  $G = (T, N, S, R, P)$

- $T$ : set of terminals
- $N$ : set of non-terminals
  - ▶ For NLP, we distinguish out a set  $P \subset N$  of pre-terminals, which always rewrite as terminals
- $S$ : start symbol
- $R$ : Rules/productions of the form  $X \rightarrow \gamma$ ,  $X \in N$  and  $\gamma \in (T \cup N)^*$

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- $P(R)$  gives the probability of each rule.

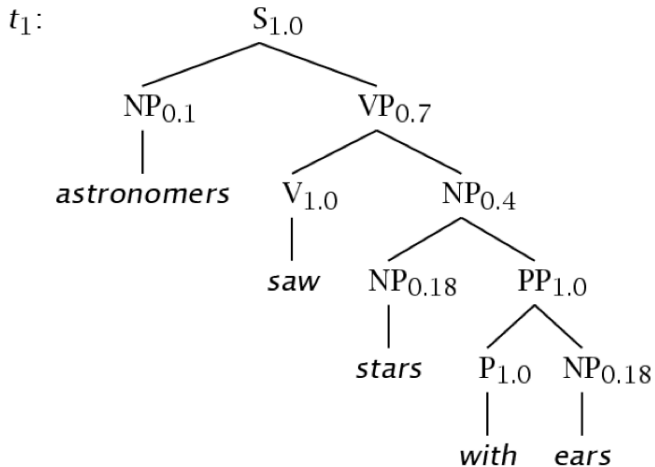
$$\forall X \in N, \sum_{X \rightarrow \gamma \in R} P(X \rightarrow \gamma) = 1$$



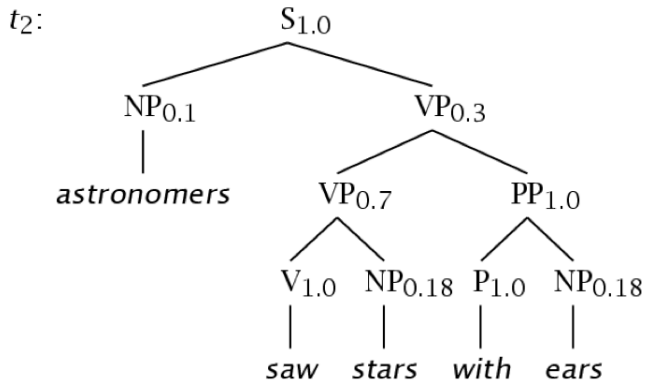
# A Simple PCFG (in CNF)

|    |   |             |     |    |   |                    |      |
|----|---|-------------|-----|----|---|--------------------|------|
| S  | → | NP VP       | 1.0 | NP | → | NP PP              | 0.4  |
| VP | → | V NP        | 0.7 | NP | → | <i>astronomers</i> | 0.1  |
| VP | → | VP PP       | 0.3 | NP | → | <i>ears</i>        | 0.18 |
| PP | → | P NP        | 1.0 | NP | → | <i>saw</i>         | 0.04 |
| P  | → | <i>with</i> | 1.0 | NP | → | <i>stars</i>       | 0.18 |
| V  | → | <i>saw</i>  | 1.0 | NP | → | <i>telescope</i>   | 0.1  |

# Example Trees



# Example Trees



# Probability of trees and strings

- $P(t)$ : The probability of tree is the product of the probabilities of the rules used to generate it
- $P(w_{1n})$ : The probability of the string is the sum of the probabilities of the trees which have that string as their yield

# *Tree and String probabilities*

## Tree and String probabilities

$w_{15}$  = *astronomers saw stars with ears*

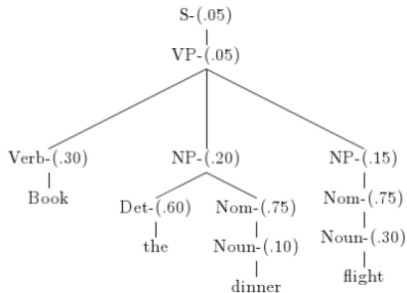
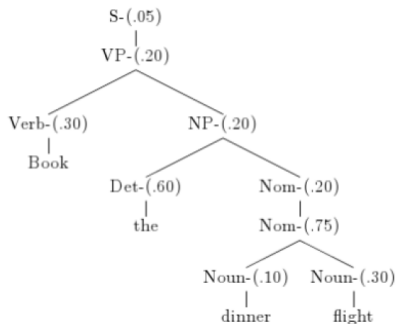
$$\begin{aligned}P(t_1) &= 1.0 * 0.1 * 0.7 * 1.0 * 0.4 * 0.18 \\&\quad * 1.0 * 1.0 * 0.18 \\&= 0.0009072\end{aligned}$$

$$\begin{aligned}P(t_2) &= 1.0 * 0.1 * 0.3 * 0.7 * 1.0 * 0.18 \\&\quad * 1.0 * 1.0 * 0.18 \\&= 0.0006804\end{aligned}$$

$$\begin{aligned}P(w_{15}) &= P(t_1) + P(t_2) \\&= 0.0009072 + 0.0006804 \\&= 0.0015876\end{aligned}$$

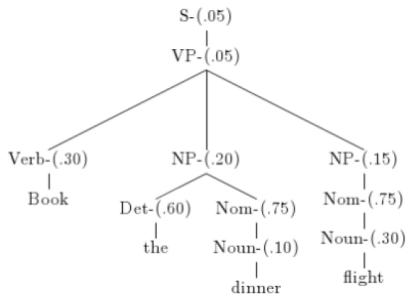
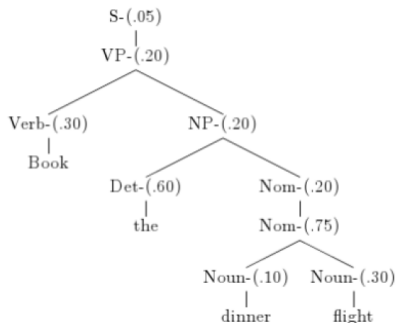
*“Book the dinner flight”*

# “Book the dinner flight”





# “Book the dinner flight”



## Probabilities

- Parse tree 1:  $.05 \times .20 \times .30 \times .20 \times .60 \times .20 \times .75 \times .10 \times .30 = 1.62 \times 10^{-6}$
- Parse tree 2:  $.05 \times .05 \times .30 \times .20 \times .60 \times .75 \times .10 \times .15 \times .75 \times .30 = 2.28 \times 10^{-7}$

# Features of PCFGs

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- Real text tends to have grammatical mistakes. PCFG avoids this problem by ruling out nothing, but by giving implausible sentences a low probability
- In practice, a PCFG is a worse language model for English than an n-gram model
- All else being equal, the probability of a smaller tree is greater than a larger tree

# Important Questions?

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- How to learn the rule probabilities in the grammar  $G$ ?